Phylogenetic analysis of a literary genre, waka, with BERT reveals mean-reverting self-excitation

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Culture evolves over time. To understand the evolution of culture, it is necessary to answer questions such as “Can culture be interpolated or extrapolated?”, “Is the time series of culture distinguishable from its reverse, or, is the arrow of time observable?”, “Is the evaluation of the influence of cultural products on later ones consistent throughout time?”, “How strong is the Matthew effect, *i.e.*, the rich-get-richer effect?”, and “What rules the evolution of culture?” To answer these questions, the neural language model BERT was trained with the corpus of *waka*, a genre of Japanese poetry (Tanaka 2025). *Waka*, specifically, *tanka*, consists of five lines with 5-7-5-7-7 syllables and is the most major poetic form in Japanese literature. The imagery and vocabulary of *waka* greatly influenced *nō*, *haiku*, and novels such as *The Tale of Genji*. The *waka* dataset created by Katsuhiro Seta and maintained by the International Research Center for Japanese Studies was used (https://lapis.nichibun.ac.jp/waka/index_era.html).

A phylogenetic tree of poems was constructed from the vector representation of each poem generated by BERT. The phylogenetic network was constructed by regarding each poem as a node in a directed network. The parent poem of a given poem is the poem closest to it among the poems older than it. The validity of the estimated parent–child relationship of the phylogenetic tree was examined by comparing it with the *honkadori* relationship. In the *honkadori* (allusive variation) relationship, a poem borrows phrases from its original poems. The parent–child relationship of the reconstructed phylogenetic network was compared with the *honkadori* relationship, described in a critical edition of an anthology. The parent–child relationship showed statistically significant agreement with the *honkadori* relationship. The result was robust to changes in the model parameter values of BERT.

The phylogenetic tree could be distinguished from a phylogenetic tree constructed from the time-reversed and shuffled data. Specifically, a value defined by the distance between the parent and child differentiates the real phylogenetic tree from the time-reversed and shuffled ones. This value suggests that the poems were not transitioning from one style to another but diverging from old to new styles. Two anthologies from different periods could be interpolated but could not be extrapolated. This result suggests that random walk plays a large part in the evolution of *waka*. The number of children of a poem on the phylogenetic tree constructed with a limited training set was positively correlated with that trained with the whole dataset. In other words, the number of children of a poem, the proxy variable of its cultural influence on later poems, evaluated at a time point, was positively correlated with that evaluated later. Poems selected for an authoritative anthology (Imperial anthology) tended to have more children than non-selected poems, which indicates the existence of the Matthew effect. Selected poems tended to have 10–50% more children than non-selected ones. These results suggest that culture evolves through the Matthew effect and random walks. A simple mathematical model that qualitatively reproduces these results is also presented.

Bibliography

Tanaka, Takuma (2025): “Mean-reverting self-excitation drives evolution: phylogenetic analysis of a literary genre, *waka*, with a neural language model.”, in: *Humanities and Social Sciences Communications* 12: 394. https://doi.org/10.1057/s41599-025-04714-1